Mechanical Fasteners

Next >

Learning Objectives

3.2.5 Joining and assembly

Subject content	Additional information	Mathematical understanding
Permanent and temporary methods: • rivets • threaded fastenings • soldering (soft and hard) • brazing • welding.		

Mechanical Fastenings

Mechanical fastenings are widely used in engineering because:

- *They provide a non-permanent method of fixing*
- *Pieces can be taken apart, or disassembled.*
- *Fastenings give additional strength/mechanical strength*
- *No need to clean/flux the pieces*
- *No requirement to heat the pieces*
- *Unlike heat (welding / hard soldering / brazing) processes mechanical fastenings will not cause heat distortion*

Next >

Thread Cutting

Next >

Internal Thread *(Female Thread)*

An ***internal thread***, also known as a ***female thread***, is a ***threaded hole*** or ***bore*** where a ***screw*** or ***bolt*** can be ***inserted*** and ***tightened***.

Think of it as the ***inside of a nut***. It's the part that provides a mating interface for an external thread (the part that screws into it).



Next >

Internal Thread *(Female Thread)*

To ***cut*** an ***internal screw thread***, a ***hole*** has to be ***drilled first*** and then a tool called a ***TAP*** is used to *cut a thread within the hole*.

TAPS are made from ***high speed steel (HSS)***. The top of the tap is square which enables the tap to be held securely in a ***TAP WRENCH***.



Next >

Internal Thread *(Female Thread)*

Taps are generally available in sets of three and are used in the following order:-

1. **Taper Tap:** This tap has much smaller teeth at the bottom than the Second or the Plug taps which allows it to get started by making a shallower thread cut



2. **Second Tap:** This tap follows the taper and has slightly more teeth and cuts the thread slightly deeper.



3. **Plug Tap:** This is the final tap and has fully formed thread cutting teeth almost to the end.

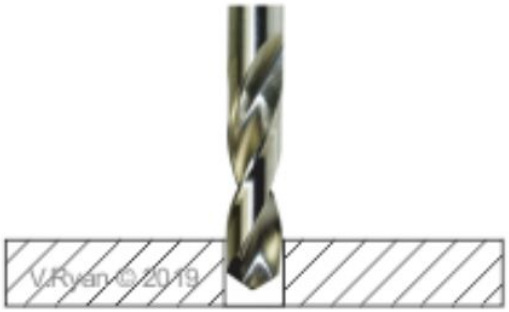


Internal Thread *(Female Thread)*

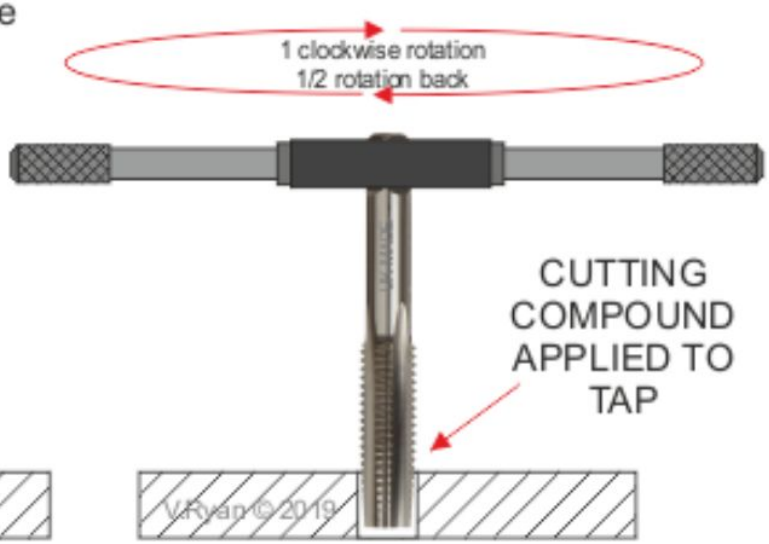
The taper tap shown below, has a long taper at the 'end' and the rest of the thread is parallel.



1. A hole is drilled through the steel, to the correct tapping size. In this case, an M8 threaded hole is required and as such, a 6.8mm diameter hole has been drilled, using a machine drill.



2. The taper tap is used first. The long taper at the end of this tap, means that it is easier to start the clockwise turning motion, that cuts the thread. If a thin piece of steel is being 'threaded', this may be the only tap that is needed.



https://technologystudent.com/equip_flsh/taps1.html

Next >

External Thread *(Male Thread)*

An ***external thread*** is a screw thread on the ***outside*** of a ***cylinder*** or ***cone***, like those found on ***bolts, screws, or studs***.

It's the ***male*** counterpart to the ***internal thread*** found *inside a nut or a tapped hole*.

An ***external thread***, also known as a ***screw thread***, is ***cut by hand*** (there are other methods) with a ***round die*** sometimes called a ***split die*** which is fixed in a ***die stock***.

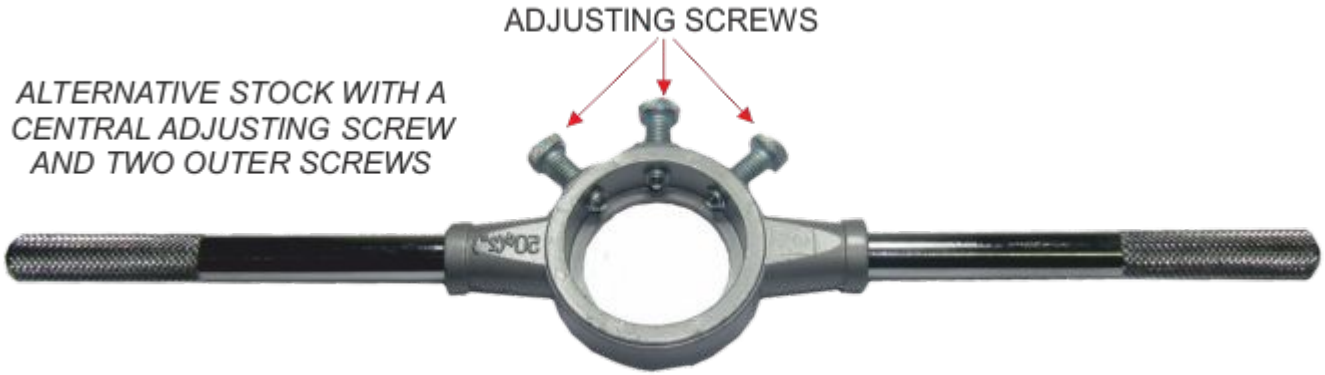


Next >

External Thread *(Male Thread)*

A ***split die*** is a threading tool used to cut external threads on a rod or shaft. It's distinguished by a split in its circumference, allowing for adjustment of the thread's fit. This split enables precise control over the thread's diameter, making it suitable for both initial cutting and fine-tuning.



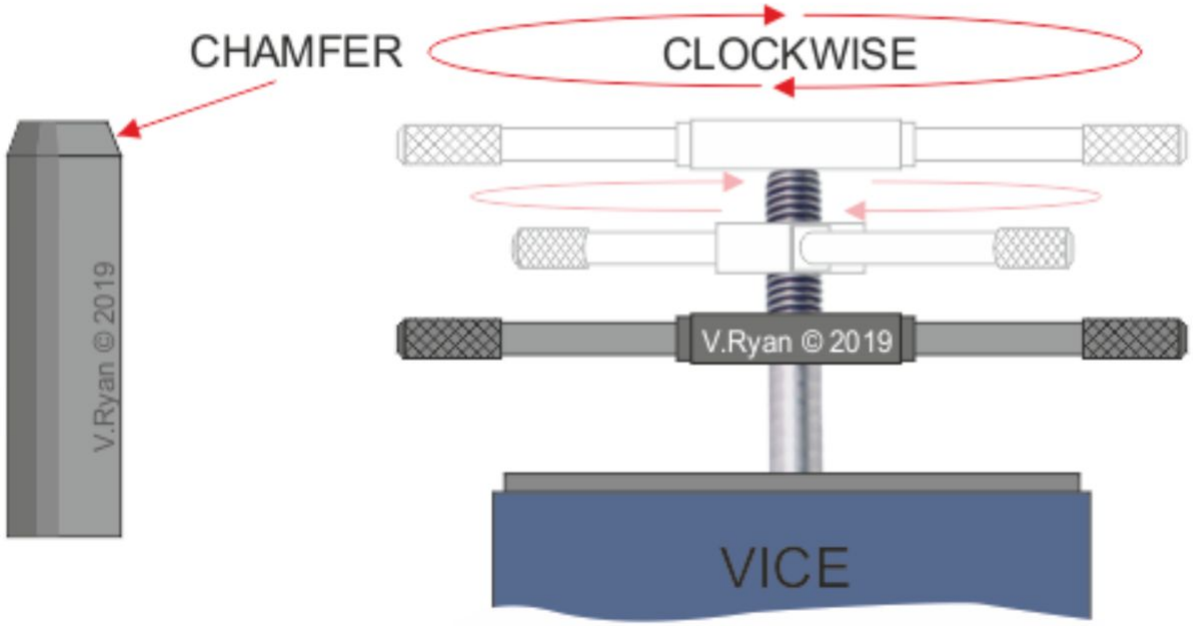


https://www.technologystudent.com/equip_flsh/exthread1.html

Next >

External Thread *(Male Thread)*

A ***die*** is used to cut or form the male portion of the mating pair (e.g. a bolt). The process of cutting or forming threads using a tap is called tapping, whereas the process using a die is called threading.





Next >

Fasteners

- Screw fastening and rivets are the most widely used semi-permanent fastenings in engineering products.
- Screwed fastenings on covers and internal components allow access and removal for maintenance and repair.



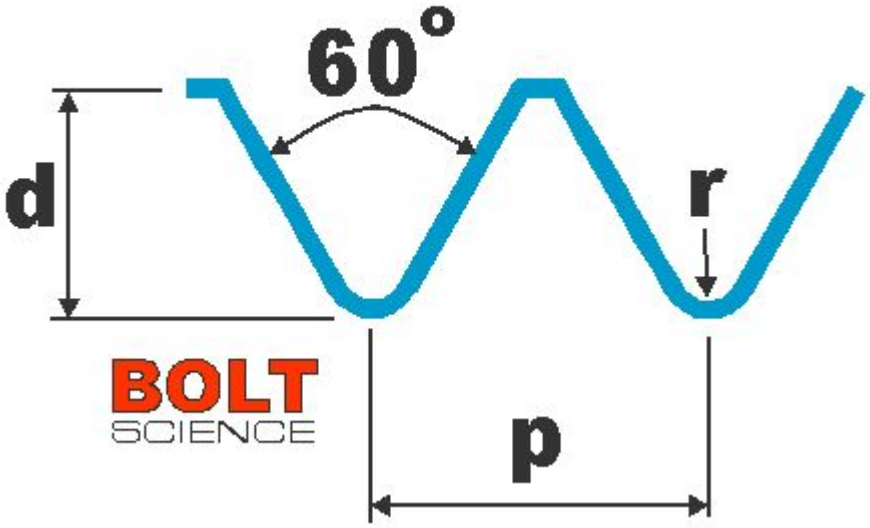
Fasteners

- Most countries throughout the world now use the International Organisation for Standardisation (ISO) metric screw thread.
- The USA is an exception where feet and inches are still in widespread use.
- Small sizes screw fastenings, particularly those used in electrical and IT equipment, employ the British Association (BA) screw thread.

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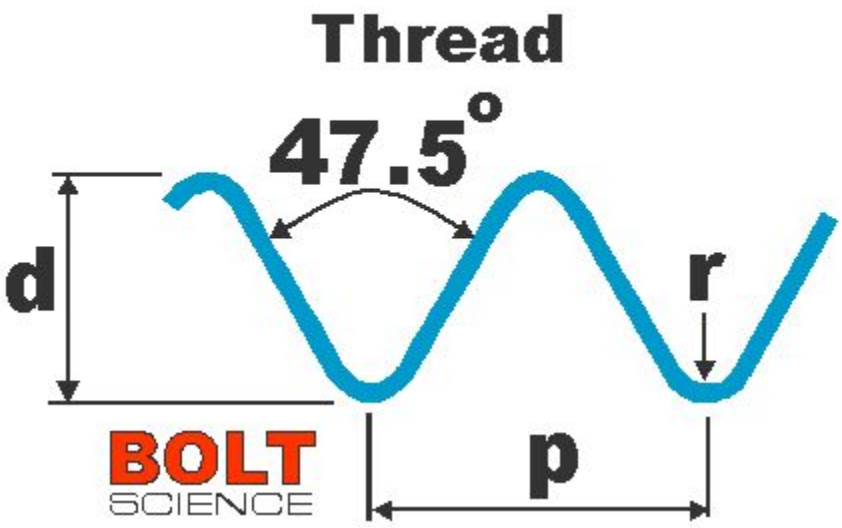
Thread Forms

Metric Thread



d = 0.866 x P (pitch)

British Association Thread



d = 0.6 x P (pitch)

Next >

Task 1a

Using the camera on your mobile phone ***‘safely’*** *walk* around the ***workshop*** – *observe and take photos* of the ***different*** *types of mechanical fasteners* used.

Mechanical Fasteners

Bolts, Studs and Set Screws

Next >

Fasteners

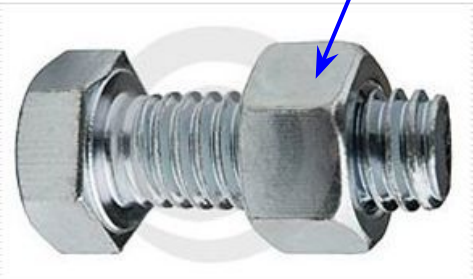
- The most commonly used *screwed fasteners are nuts and bolts, setscrews, studs and self tapping screws*.
- Metric nuts and bolts are specified in a way that describes their shape of head and nut, nominal diameter, pitch and bolt length.

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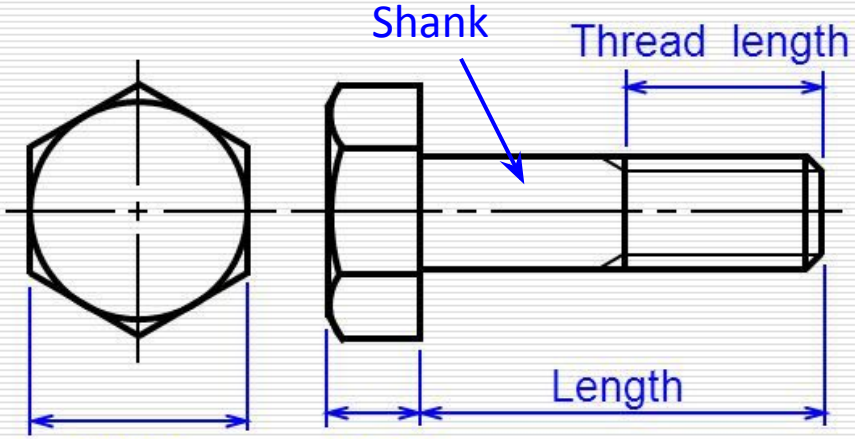
Hexagonal Head Nut and Bolt

BOLT : Terminology

Bolt is a threaded cylinder with a head.



Hexagonal head bolt and nut



Hexagonal Head Nut and Bolt

A *hexagonal head nut and bolt* might be specified as follows:

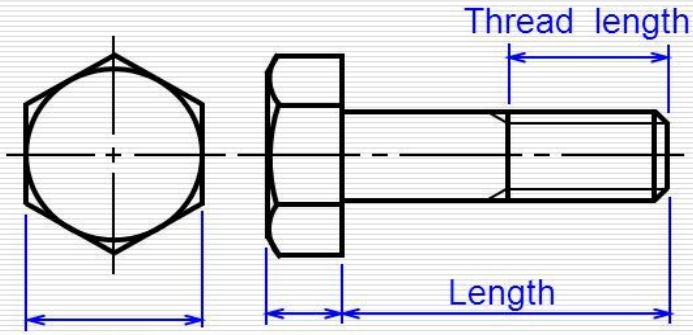
- ***Steel, hex hd bolt:*** *M12 x 1.25 x 100*
- ***Steel, hex hd nut:*** *M12 x 1.25*

BOLT : Terminology

Bolt is a threaded cylinder with a head.



Hexagonal head bolt and nut



Hexagonal Head Nut and Bolt

- Bolts can be obtained with different lengths of thread for different applications.
- bolts for general use are ‘forged’ to shape and are generally known as ***‘black bolts’***.
- For applications where precision is required, fitted bolts can be obtained that have been accurately machined to size.
- Bolts made from ***‘high tensile steel’*** are also supplied for heavy duty applications.

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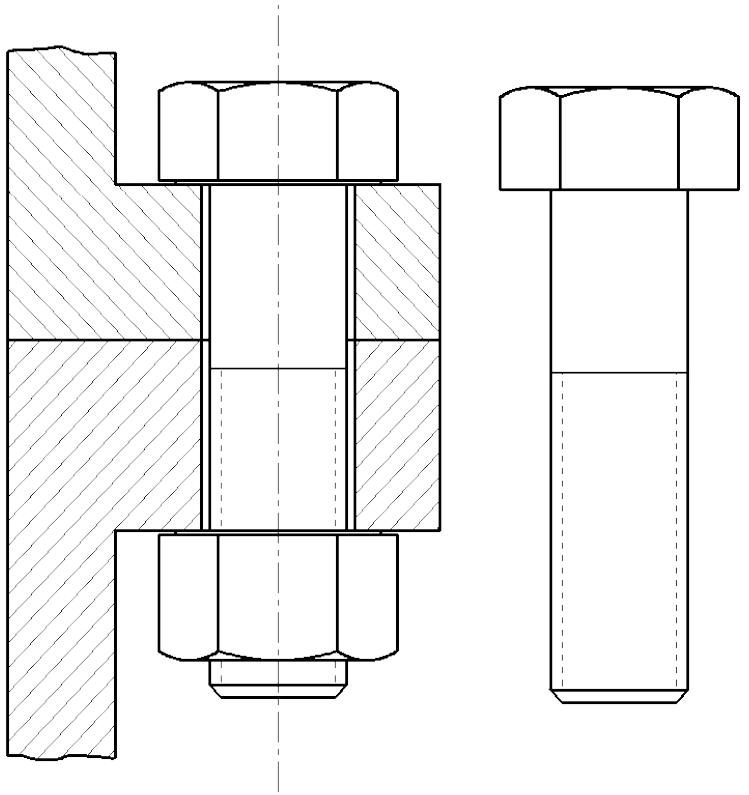
Hexagonal Head Nut and Bolt

- Nuts and bolts are made from steel, brass, aluminium alloys and plastics.
- Steel nuts and bolts are sometimes cadmium plated to improve their resistance to corrosion.

Next >

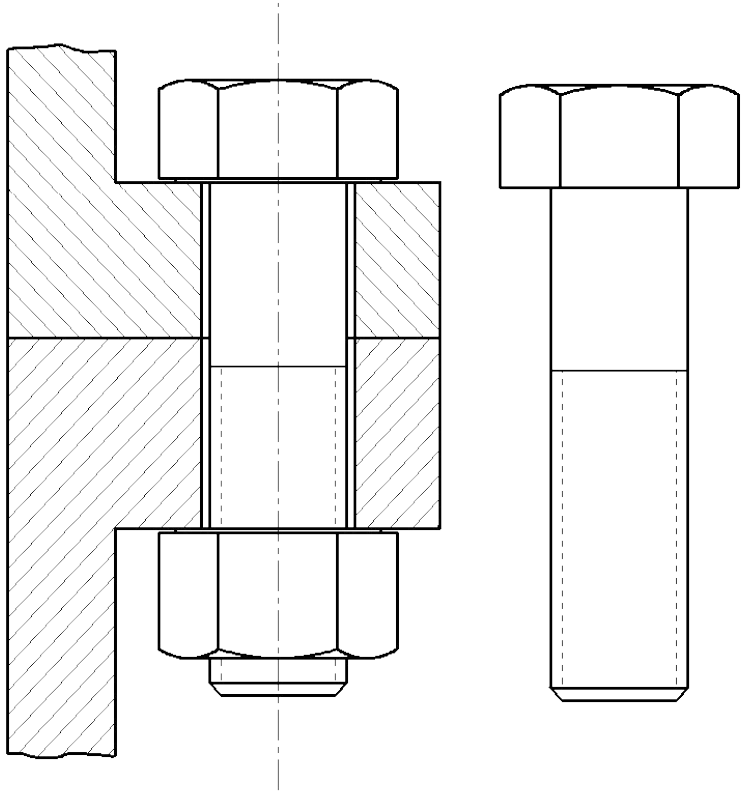
Hexagonal Head Nut and Bolt

- The diagram shows two components which have been joined by a nut and bolt.
- A bolt should always be selected such that the plain unthreaded part of its shank extends through the joint surface.



Hexagonal Head Nut and Bolt

- This is to ensure that there will be no ***shearing force*** on the *weaker threaded part*.
- It is ***<u>good practice</u>*** to place a ***'washer'*** under the ***nut*** – this *prevents damage to the component face and has the effect of spreading the load*.



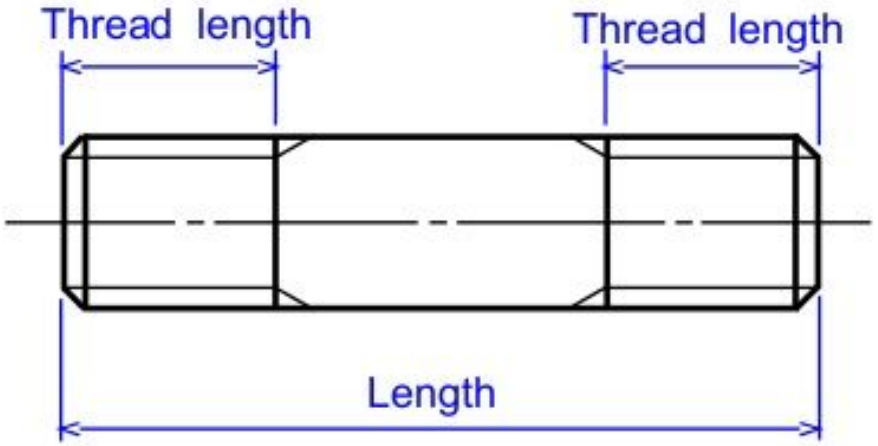
Stud

STUD : Terminology

Stud is a *headless* bolt, threaded at both ends.

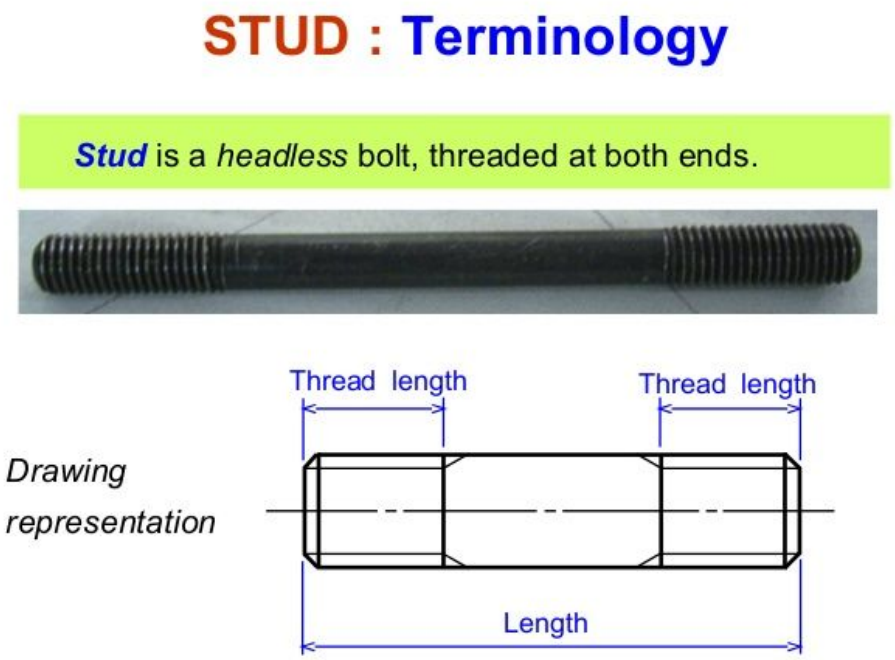


Drawing representation



Stud

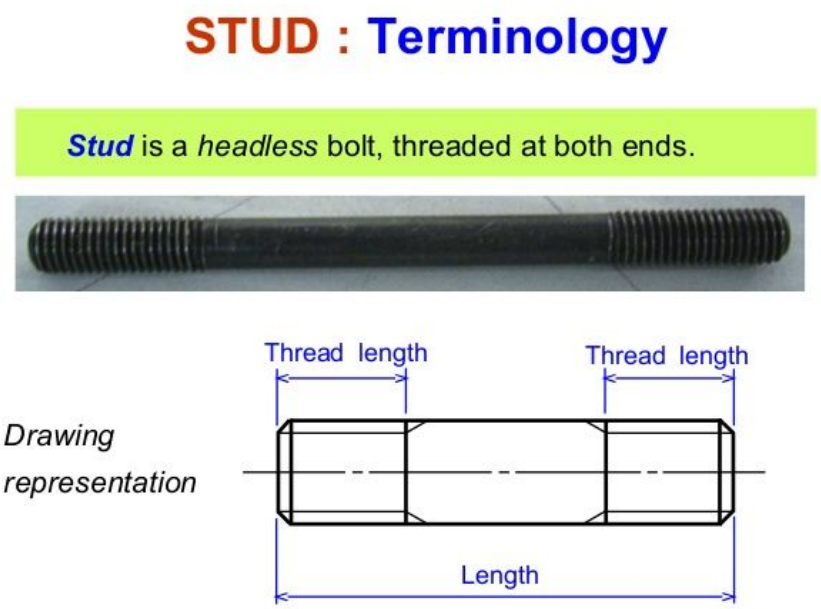
- A stud is a length of bar which has been threaded at each end.
- One (usually the shorter) end is screwed into the major component – sometimes using a locking fluid to prevent it from easily becoming unscrewed.



Next >

Stud

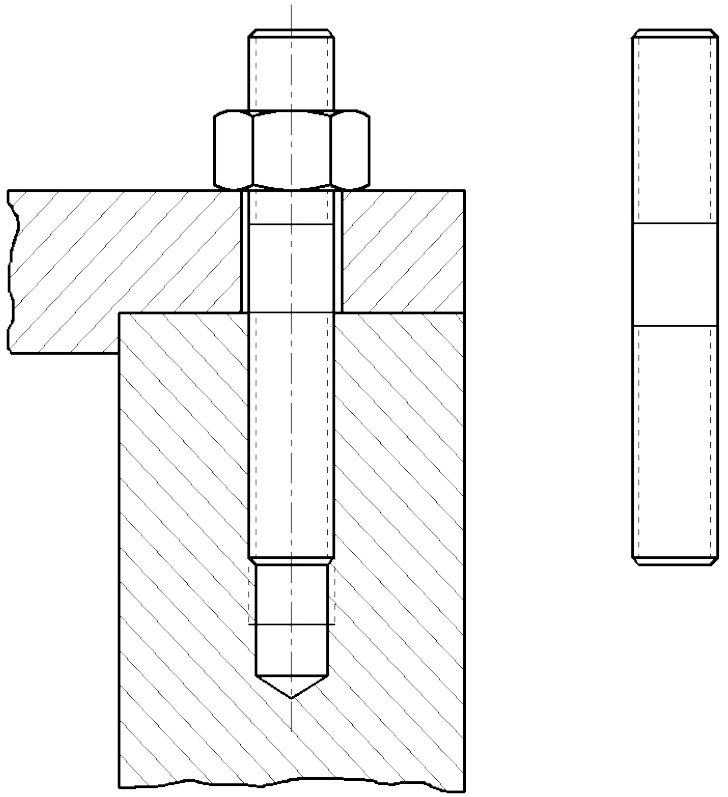
- It is good practice to use studs where a component such as a machine inspection cover needs to be moved regularly for maintenance.
- Because the studs are permanently in position in the machine casing – wear to the threaded holes is prevented



Next >

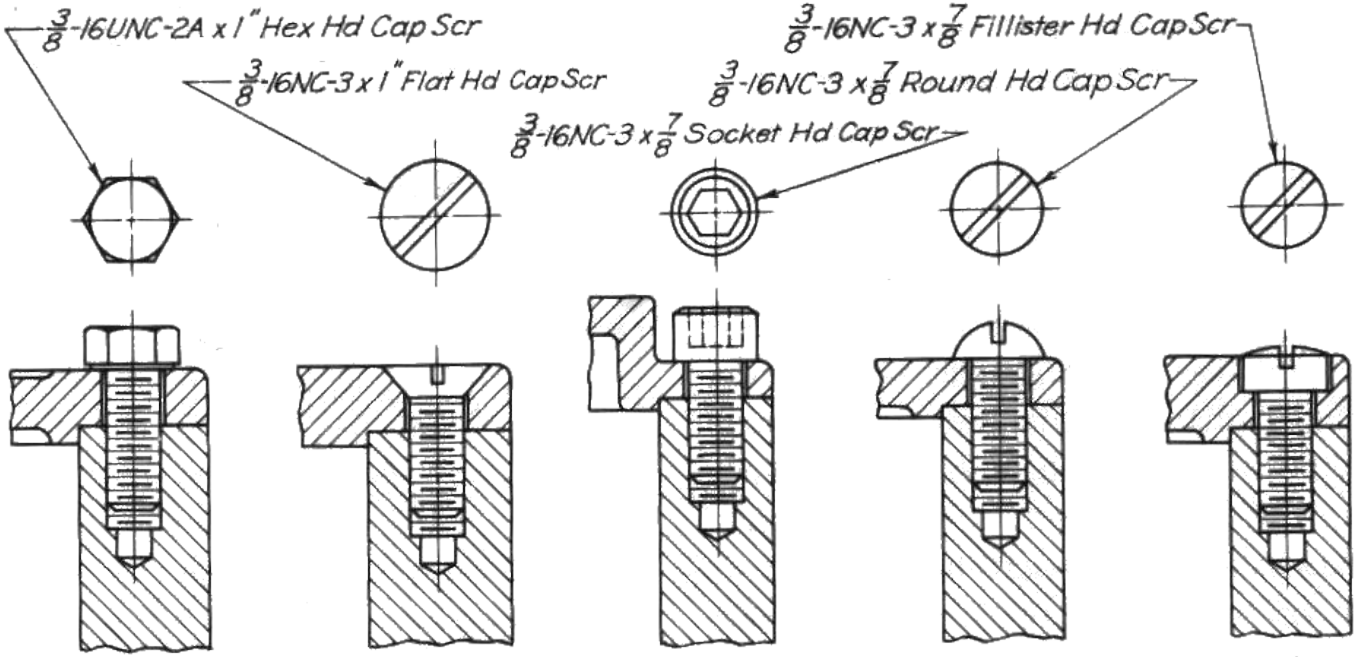
Stud

- The unthreaded portion, in the middle of a stud should be a little shorter than the component through which it passes, and the threaded nut end should protrude through by a distance greater than the thickness of the nut.
- It is also good practice to place a washer under the nut.



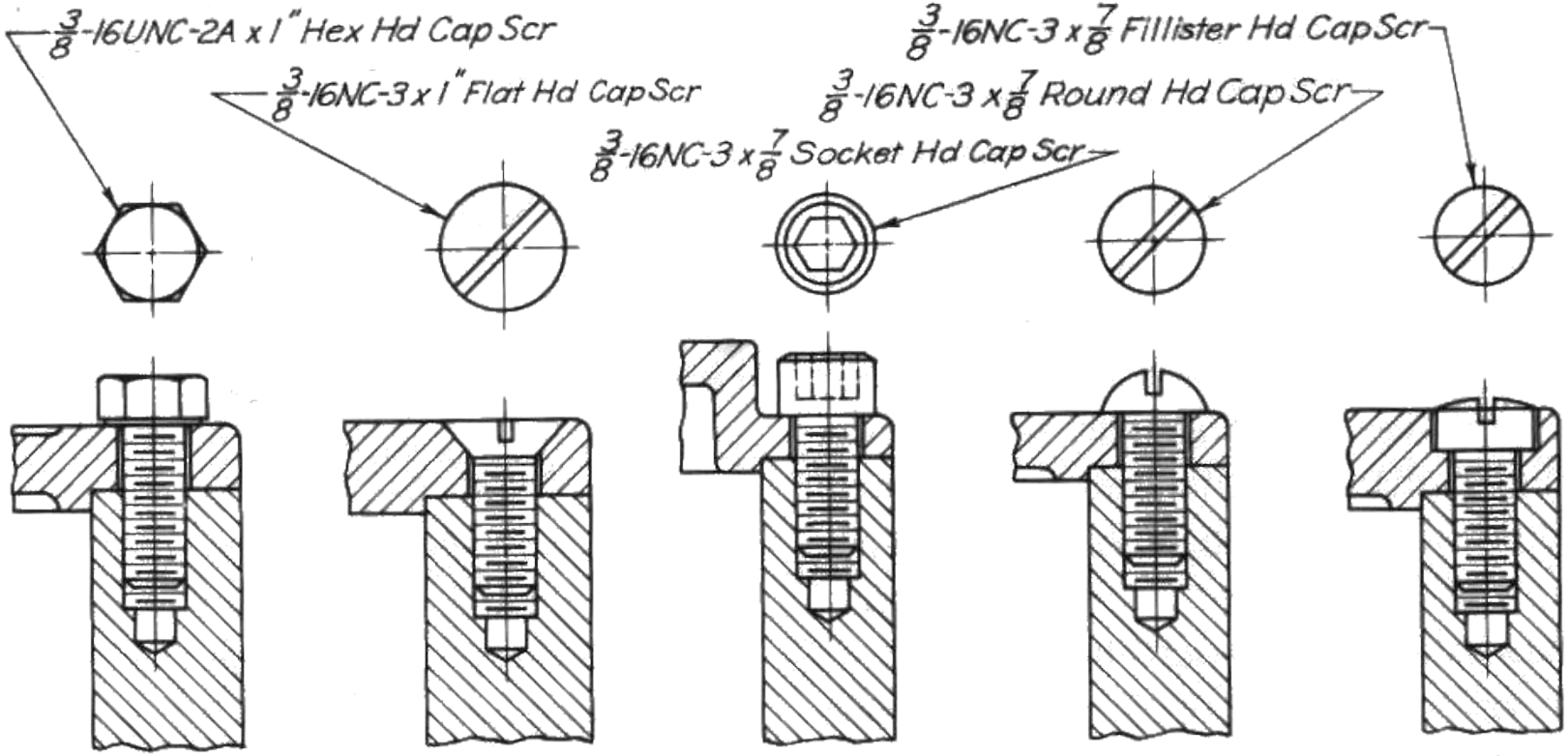
Set Screws / Machine Screws

- ***Set Screws*** – also called ***machine screws*** – do ***<u>not</u>*** use a ***<u>nut</u>***.
- They are usually used when joining sheet metal or thin plate components to a bulky object it is usual to use screws which are threaded over the whole of the shank.



Next >

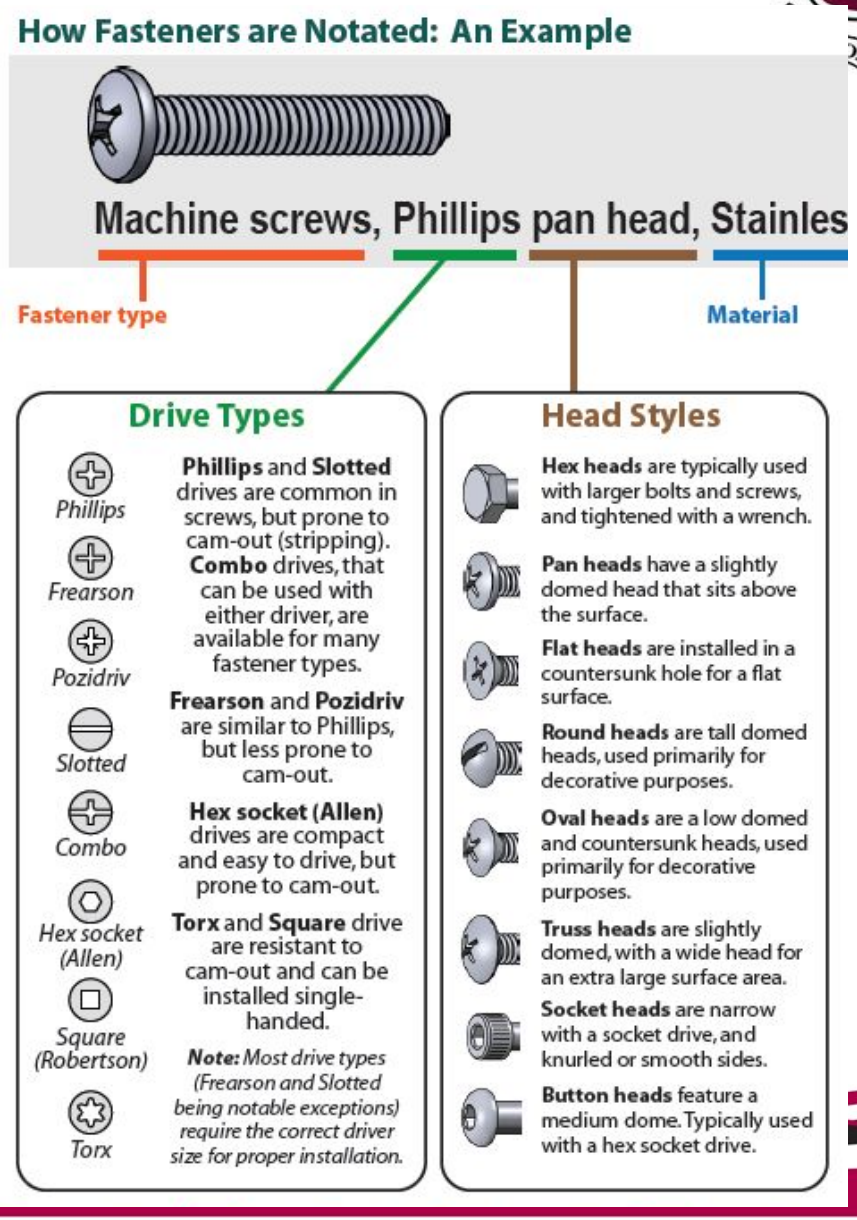
Set Screws / Machine Screws



Next >

Set Screws/Machine Screws

- Machine Screws are defined by the drive type, head style, material and size (e.g. M5 x 0.6)



Next >

Locking Devices
Nuts and Washers

Next >

Locking Devices

Screwed fastenings that are subject to vibration require a locking device to prevent them working loose.

There are two basic categories of locking device:

- *Those with 'positive' locking action*
- *Those that depend on 'friction'*

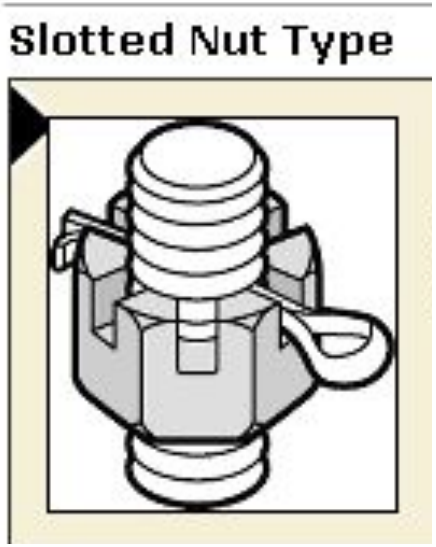
Next >

Positive Locking Devices

Includes:

Castle nuts – secured with a split pin

- *The split pin passes through a hole which is drilled in the bolt and its ends are opened to ensure that it stays in position.*
- *Split pins are only used once*







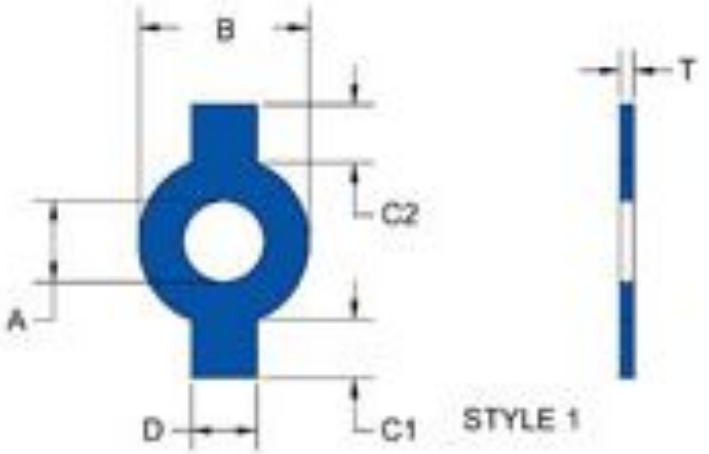
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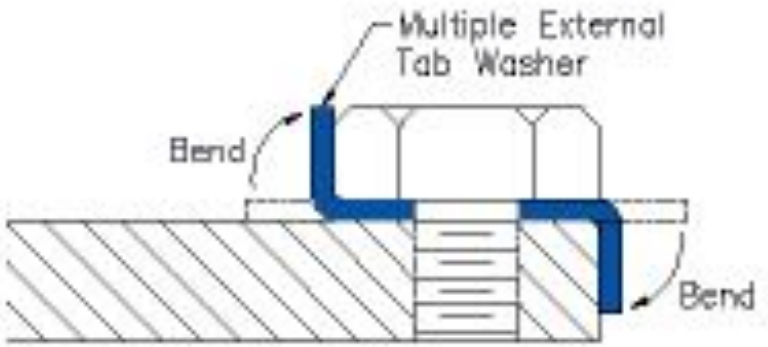
Positive Locking Devices

Tab Washers

- *After tightening the nut – one side is bent up against the nut face and the opposite side is bent over the component as shown.*

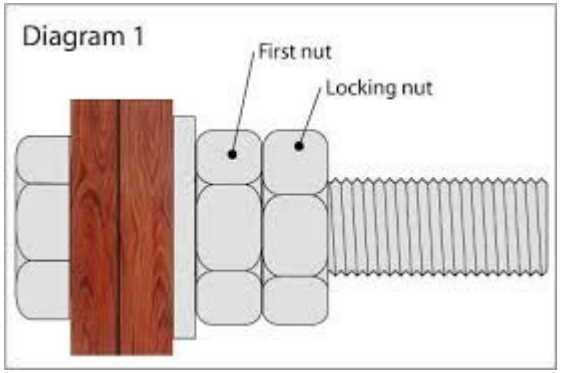
Example: Tab Washer



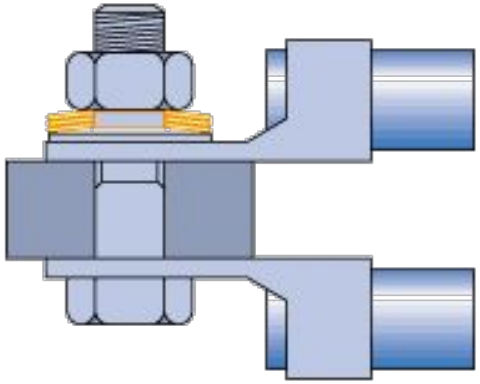


Fiction Locking Devices

- *Locking devices which depend on friction include lock nuts, stiff nuts (Nyloc Nuts) and spring washers.*





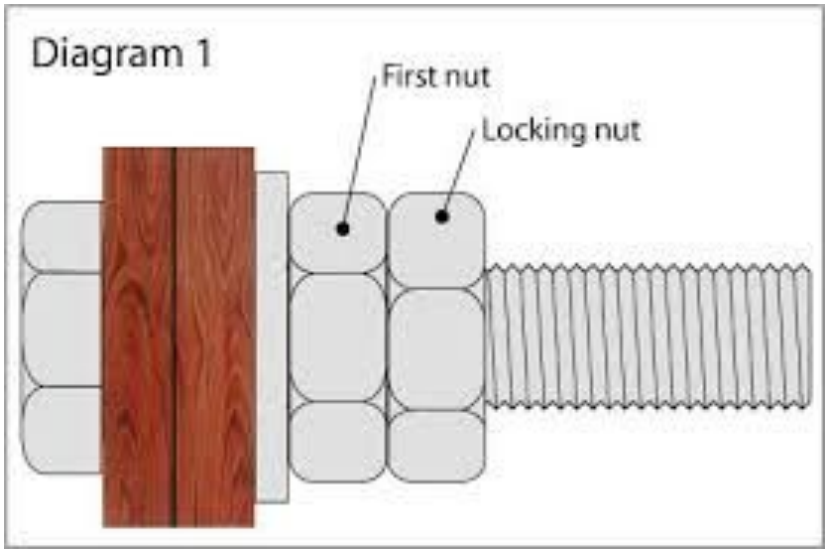


Next >

Fiction Locking Devices

Lock Nuts

A ***lock nut*** is a secondary nut (usually) of a smaller thickness that is tightened down on top of a plain nut.



The combined frictional resistance is generally sufficient to withstand the effects of vibrations.

Next >

Fiction Locking Devices

Stiff Nuts

There are a variety of types of ***stiff nuts***.

The most common types have a compressed ***fibre*** or a ***nylon insert***.

Those with a fibre insert are traditionally called ***'Simmonds' nuts.***

And those with nylon inserts are referred to as ***Nyloc nuts.***

It's good practice to replace these type of nut after maintenance.



Next >

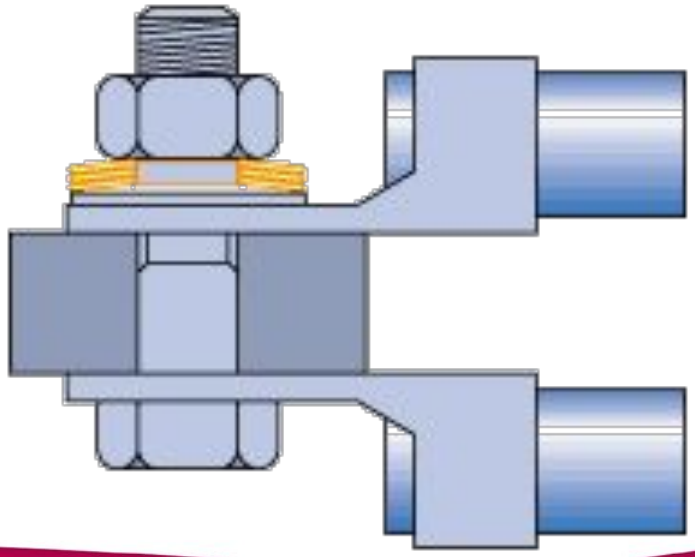
Fiction Locking Devices

Spring Washers

Spring washers come in a ***variety*** of shapes.



When the nut is tightened down on it, the edges of the split resist anticlockwise turning.



Nut Types



Hex
A six sided nut. Also referred to as a Finished Hex Nut.



Heavy Hex
A heavier pattern version of a standard hex nut.



Nylon Insert Lock
A nut with a nylon insert to prevent backing off. Also referred to as a Nylock.



Jam
A hex nut with a reduced height.



Nylon Insert Jam Lock
A nylock nut with a reduced height.



Wing
A nut with 'wings' for hand tightening.



Cap
A nut with a domed top over the end of the fastener.



Acorn
Acorn nuts are a high crown type of cap nut, used for appearance.



Flange
A nut with a built in washer like flange.



Tee
A nut designed to be driven into wood to create a threaded hole.



Square
A four sided nut.



Prevailing Torque Lock
A non-reversible lock nut used for high temperature applications.



K-Lock or Kep
A nut with an attached free-spinning external tooth lock washer.



Coupling
Coupling nuts are long nuts used to connect pieces of threaded rod or other male fasteners.



Slotted
Slotted nuts are used in conjunction with a cotter pin on drilled shank fasteners to prevent loosening.



Castle
Castle nuts are used in conjunction with a cotter pin on drilled shank fasteners to prevent loosening.

Next >

Washer Types



Flat
A flat washer, used to distribute load. Available in SAE, USS and other patterns.



Fender
An oversize flat washer used to further distribute load especially on soft materials.



Finishing
A washer used to obtain a 'finished' look. Usually used with oval head screws.



Split Lock
The most common style of washer used to prevent nuts and bolts from backing out.



External Tooth Lock
A washer with external 'teeth'. Used to prevent nuts and bolts from backing out.



Internal Tooth Lock
A washer with internal 'teeth'. Used to prevent nuts and bolts from backing out.



Square
A square shaped washer.



Dock
Dock washers have a larger outside diameter and are thicker than standard.



Ogee
Thick, large diameter, cast iron washers with a curved or sculpted appearance. Typically used in dock and wood construction.



Sealing
A soft neoprene washer bonded to a metal backing. Used to seal out air/water or dampen noise and vibration.

Next >

Self Tapping Screws

Next >

Self Tapping Screws

Self tapping screws are widely used with sheet metal and plastic components.

The type used depends on the hardness of the receiving material.

For each type, a pilot hole is drilled whose diameter is the same or slightly less than the ***'root'*** diameter of the thread.

Next >

1. Thread Forming Screws

The thread forming type is used on soft materials such as thermoplastics.

As it is screwed home it displaces the material to form its own mating thread.



Next >

2. Thread Cutting Screws

The thread cutting type is used on harder materials.

It has a cutting edge at the start of its shank.

This cuts a thread in the receiving material, rather like a screw cutting tap



Next >

Task 1b

Using the photos you took showing how different *mechanical fasteners* have been used to secure applications in the workshop and around the UTC.

Produce a ***fact sheet*** showing:

1. *The application being secured by the mechanical fastening*
2. *Stating which mechanical fastening has been used and why it has been used – including estimated size (e.g. M10), material, drive type, head style and any additional components e.g. nuts and washers*
3. *Your fact sheet should show a diagram/image of the mechanical fastening on its own*